The Méndez–Vaz Pinto–Villarreal Conjecture: Counterexamples and Diagonal Representability

Abstract

We study the Méndez–Vaz Pinto–Villarreal conjecture that a Simis monomial ideal with no embedded primes and pairwise distinct irreducible-component radicals is a standard diagonal weighting of a squarefree Simis ideal. The family $I_{r,s} = (ac, bd, abz^r, cdz^s)$ satisfies these hypotheses for all positive integers r, s, but has such a representation exactly when $r = s$. Multiplication by a new variable gives counterexamples with contractible support complexes. We prove that the conjectured conclusion holds if the generator supports agree with those of the radical and contain no strong triangle. The argument combines known symbolic-square obstructions with a combinatorial isolation lemma. A second criterion obtains diagonal representability from $I^{(2)} = I^2$ when the union of any two intersecting component supports contains at most one further component support; the counterexample family shows this bound is sharp. Even-cycle and disjoint-block families describe the scope and limitations of these criteria.

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1 Introduction

Let $S = k[x_1, \ldots, x_n]$ be a polynomial ring over a field. All ideals considered below are monomial. We use symbolic powers with respect to minimal primes:

$$I^{(q)} = \bigcap_{P \in \mathrm{Min}(I)} (I^q S_P \cap S), \qquad q \geq 1.$$

An ideal is *Simis* if $I^{(q)} = I^q$ for every $q \geq 1$. For a positive integer vector d, the standard diagonal weighting is induced by $x_i \mapsto x_i^{d_i}$; it is denoted I^{δ_d}.

Méndez, Vaz Pinto, and Villarreal conjectured that a nonzero proper Simis monomial ideal without embedded primes, whose irredundant irreducible decomposition has pairwise distinct radicals, is of the form J^{δ_d}, where $J \subset S$ is squarefree and Simis [MPV25, Conjecture 5.7]. The representation is in the original polynomial ring. We call the three assumptions just listed the *MPV hypotheses*, and call the asserted representation the *MPV conclusion*. Pairwise distinct radicals are essential: an irredundant irreducible decomposition need not have this property.

The first result below disproves the general implication. For every $r, s > 0$, the ideal

$$I_{r,s} = (ac, bd, abz^r, cdz^s) \subset k[a, b, c, d, z]$$

is Simis and has four irreducible components with distinct, incomparable radicals. Its generator supports also agree with those of its radical. Nevertheless, a squarefree diagonal root exists if and

only if $r = s$ (Theorem 3.1). The obstruction survives multiplication by a new variable, although the resulting support complex is contractible (Corollary 3.3). Thus support preservation and the condition $\beta_1 \leq 1$ do not repair the conjecture.

Our principal positive result is Theorem 4.1: the MPV conclusion holds if the MPV hypotheses are supplemented by support preservation and the absence of a strong triangle in the generator-support clutter. In particular it holds for balanced support clutters. The new combinatorial step is Lemma 4.3, which produces an isolated pair from any nonconstant labeling of the facets through a vertex. Combined with the local exponent obstructions of Bordoloi–Das–Kumar, it forces a single positive occurrence exponent for each variable.

Theorem 5.2 gives a second criterion, read from irreducible-component supports. If the union of two intersecting supports contains at most one further component support, equality of the second symbolic and ordinary powers forces a squarefree diagonal root. The bound cannot be increased to two, even for all-power Simis ideals. The two criteria concern different incidence data: generator supports in the first, component supports in the second.

Relation to previous work. The basic weighting and symbolic-power formulas are due to Méndez–Vaz Pinto–Villarreal [MPV25]. Bordoloi–Das–Kumar prove the conjecture for exact support-2 ideals [BDK26b], and for exact support-3 ideals and simplicial forests [BDK26a]. Here *exact support* p means that every minimal generator involves exactly p variables. Their one-generator-per-support lemma and isolated-pair proposition are recalled explicitly in Section 4; we do not regard those algebraic ingredients as new. The no-strong-triangle criterion extends the forest route, but does not include every exact support-3 case.

Bijender–Kumar give a criterion involving exponent-mismatching component pairs and establish the height-two case [BK25]. Section 5.1 explains the relationship precisely. Once diagonal representability is known, their mismatch condition is vacuous. Consequently our represented examples do not separate the successful class of their theorem. The contribution here is the support criterion that establishes representability, together with the component obstruction and its sharp bound.

Section 6 supplies an infinite support-4 family outside the named forest and exact-support-3 classes, and a disjoint-block family showing that the absence of strong triangles is not necessary. Its squarefree Simis assertions use the established balanced-to-Mengerian implication and the symbolic-power characterization in [FHO74, HHTZ08].

2 Notation and diagonal roots

Write $\mathcal{G}(I)$ for the minimal monomial generators, and set

$$\mathcal{F}(I) = \{\operatorname{supp}(u) : u \in \mathcal{G}(I)\}, \qquad \Delta(I) = \langle \mathcal{F}(I) \rangle.$$

A *clutter* is an antichain of nonempty subsets of a finite vertex set. Support preservation, $\mathcal{F}(I) = \mathcal{F}(\sqrt{I})$, means that $\mathcal{F}(I)$ is a clutter, although several generators may initially have the same support. For a clutter $\mathcal{H}$, its *blocker* $b(\mathcal{H})$ consists of its inclusion-minimal vertex covers; $b(b(\mathcal{H})) = \mathcal{H}$.

A *strong q-cycle*, $q \geq 3$, consists of distinct vertices $v_1, \ldots, v_q$ and distinct edges $E_1, \ldots, E_q$ with

$$E_i \cap \{v_1, \ldots, v_q\} = \{v_i, v_{i+1}\},$$

where indices are cyclic. A clutter is *balanced* if it has no strong odd cycle. A facet F is a leaf if it is the only facet, or there is another facet G such that $F \cap H \subseteq F \cap G$ for every other facet H. A *simplicial forest* is a complex every nonempty facet-generated subcomplex of which has a leaf [Far02]. Subcomplexes used in this definition are generated by subsets of the facet set.

Every nonzero proper monomial ideal has a unique irredundant irreducible monomial decomposition, up to order,

$$I = \bigcap_{j=1}^{m} Q_j, \qquad Q_j = (x_i^{a_{j,i}} : i \in A_j), \qquad P_j = \sqrt{Q_j} = (x_i : i \in A_j). \tag{1}$$

The sets A_j are nonempty and the exponents positive. Each Q_j is P_j-primary, and $\operatorname{Ass}(S/I) = \{P_j : 1 \le j \le m\}$. If the radicals are distinct and there are no embedded primes, they are incomparable and

$$I^{(q)} = \bigcap_{j=1}^{m} Q_j^q \qquad (q \ge 1). \tag{2}$$

More generally one first groups irreducible components with the same radical, and uses the resulting primary components at minimal primes [MPV25, Section 1].

For $d \in \mathbb{Z}^n_{>0}$, set $\delta_d(\alpha) = (d_1\alpha_1, \ldots, d_n\alpha_n)$. Positive diagonal weighting preserves divisibility and satisfies

$$x^\beta \in L^{\delta_d} \iff x^{\lfloor \beta/d \rfloor} \in L. \tag{3}$$

It therefore commutes with finite intersections and preserves and reflects inclusions. Its minimal generators are the weighted minimal generators of L. We use two further results from [MPV25, Corollaries 4.6 and 5.5(b)]: weighting preserves the canonical primary decomposition and associated primes, and

$$L \text{ is Simis} \iff L^{\delta_d} \text{ is Simis.} \tag{4}$$

The latter equivalence requires no preliminary no-embedded-prime assumption.

Proposition 2.1 (Diagonal-root criteria)**.** *Let I be a nonzero proper monomial ideal with decomposition (1). Define*

$$D_i = \{\alpha_i > 0 : x^\alpha \in \mathcal{G}(I)\}, \quad C_i = \{a_{j,i} : i \in A_j\},$$

and let X_I consist of pairs (k, ℓ), $k < \ell$, for which $a_{k,i} \neq a_{\ell,i}$ at some $i \in A_k \cap A_\ell$. Then $C_i = D_i$, and the following are equivalent:

(i) $I = J^{\delta_d}$ for a squarefree $J \subset S$ and $d > 0$;

(ii) $|D_i| \le 1$ for every i;

(iii) $|C_i| \le 1$ for every i;

(iv) $X_I = \varnothing$.

When these conditions hold, $J = \sqrt{I} = \bigcap_j P_j$ is the unique squarefree root. Each used variable has its weight forced by its common occurrence exponent; weights of unused variables are arbitrary. The prime intersection is irredundant, so the component radicals are distinct and incomparable, and I has no embedded primes. No such structural hypotheses are needed in advance.

Proof. In a squarefree diagonal representation, each positive generator exponent at x_i is d_i. Conversely, if every D_i has at most one element, replacing each minimal generator by its squarefree support produces a squarefree ideal whose weighting is I. These supports are an antichain: containment would, by the common occurrence exponents, give divisibility of the original generators. This also proves uniqueness of the root and the stated freedom of weights.

To show $C_i \subseteq D_i$, fix $i \in A_j$. Irredundance supplies a monomial $x^\beta \in \bigcap_{h\neq j} Q_h \setminus Q_j$. Increase its exponents at all variables x_t, $t \in A_j$, to $a_{j,t} - 1$, obtaining y. Then $y \notin I$, but $x_i y \in I$. A minimal generator dividing $x_i y$ cannot divide y, so its exponent at x_i is exactly $a_{j,i}$.

Conversely, if $x^\alpha \in \mathcal{G}(I)$ and $\alpha_i > 0$, then $x^{\alpha-e_i} \notin I$. Choose Q_j not containing this monomial. Since $x^\alpha \in Q_j$, the changed coordinate must cross the threshold of Q_j, giving $\alpha_i = a_{j,i}$. Thus $C_i = D_i$; the equivalence with $X_I = \varnothing$ follows immediately.

With coherent exponents d_i, one has $Q_j = P_j^{\delta_d}$, and (3) gives $I = (\bigcap_j P_j)^{\delta_d}$. If a prime component were redundant, weighting would make the corresponding Q_j redundant. Thus the prime intersection is irredundant, which also excludes repeated or comparable radicals. The associated-prime assertion follows from (1). □

3 A counterexample and its cone

Theorem 3.1. *For positive integers r, s, let*

$$I_{r,s} = (ac, bd, abz^r, cdz^s) \subset S = k[a, b, c, d, z].$$

Its irredundant irreducible decomposition is

$$I_{r,s} = Q_1 \cap Q_2 \cap Q_3 \cap Q_4, \tag{5}$$

where

$$Q_1 = (a, b, z^s), \quad Q_2 = (a, d), \quad Q_3 = (b, c), \quad Q_4 = (c, d, z^r).$$

The radicals $P_j = \sqrt{Q_j}$ are distinct and incomparable, of heights $3, 2, 2, 3$, and

$$I_{r,s}^{(q)} = I_{r,s}^q = \bigcap_{j=1}^{4} Q_j^q \qquad (q \geq 1). \tag{6}$$

In particular $I_{r,s}$ satisfies the MPV hypotheses and

$$\sqrt{I_{r,s}} = M = (ac, bd, abz, cdz), \qquad \mathcal{F}(I_{r,s}) = \mathcal{F}(M).$$

It has a squarefree positive-diagonal root if and only if $r = s$. For $r = s = t$, the root is M and the unique weight vector, in the order (a, b, c, d, z), is $(1, 1, 1, 1, t)$. The root M is Simis. For $r \neq s$, there is no squarefree diagonal root even without requiring the root to be Simis.

Proof. Put $D_0 = S$ and $D_q = \bigcap_j Q_j^q$. A monomial $u = a^\alpha b^\beta c^\gamma d^\delta z^\zeta$ belongs to D_q precisely when

$$\alpha + \beta + \lfloor \zeta/s \rfloor \geq q, \quad \alpha + \delta \geq q, \quad \beta + \gamma \geq q, \quad \gamma + \delta + \lfloor \zeta/r \rfloor \geq q. \tag{7}$$

Indeed, membership in the power of a pure-power ideal is equivalent to having total available capacity at least q among its generators.

Each generator of $I_{r,s}$ lies in every Q_j, so $I_{r,s}^q \subseteq D_q$. For the reverse inclusion, use induction on q. If $ac \mid u$, division by ac reduces each left-hand side in (7) by one, hence $u/(ac) \in D_{q-1}$. The same holds for division by bd.

Suppose neither divides u. If $\alpha = 0$, the second inequality forces $\delta \geq q > 0$, so $\beta = 0$. The other inequalities give $\gamma \geq q$ and $\zeta \geq sq$, hence $(cdz^s)^q \mid u$. If $\alpha > 0$, then $\gamma = 0$; the third inequality gives $\beta \geq q > 0$, hence $\delta = 0$, and the remaining inequalities give $\alpha \geq q$, $\zeta \geq rq$. Thus $(abz^r)^q \mid u$. This proves $D_q = I_{r,s}^q$ for all q, including (5) at $q = 1$.

The four generators have incomparable supports. The monomials

$$cdz^{s-1}, \quad bc, \quad ad, \quad abz^{r-1}$$

belong to all but, respectively, Q_1, Q_2, Q_3, Q_4. Thus the decomposition is irredundant. Its radicals are visibly incomparable, so they are exactly the associated and minimal primes. Formula (2) now gives (6).

Taking squarefree supports gives M. The positive occurrence sets are $\{1\}$ for a, b, c, d and $\{r, s\}$ for z. Proposition 2.1 proves the representation statement and forces the weights on the diagonal. The case $r = s = 1$ of (6) proves that M is Simis. □

Remark 3.2. The variable z cannot be omitted from the last two generators of the root. If $H = (ac, bd, ab, cd)$, then $H^{\delta_{(1,1,1,1,t)}} = H \neq I_{t,t}$, since $ab \in H \setminus I_{t,t}$.

The generator and component clutters are

$$\mathcal{H} = \{\{a, c\}, \{b, d\}, \{a, b, z\}, \{c, d, z\}\},$$

$$\mathcal{A} = \{\{a, b, z\}, \{a, d\}, \{b, c\}, \{c, d, z\}\}.$$

They are mutual blockers: $b(\mathcal{H}) = \mathcal{A}$ follows from the minimal primes of M, and blocker involution gives the reverse equality. The sequences

$$(a, \{a, c\}, c, \{c, d, z\}, z, \{a, b, z\}, a)$$

and

$$(a, \{a, d\}, d, \{c, d, z\}, z, \{a, b, z\}, a)$$

are strong triangles in the two clutters. The omitted edges $\{b, d\}$ and $\{b, c\}$, respectively, avoid the cycle vertices. Thus neither clutter is balanced.

Corollary 3.3 (Contractible counterexamples)**.** *In $R = S[t]$, put $K_{r,s} = tI_{r,s}R$. Its irredundant irreducible decomposition is*

$$K_{r,s} = (t) \cap Q_1R \cap Q_2R \cap Q_3R \cap Q_4R.$$

Its associated primes are precisely the five distinct minimal primes $(t), P_1R, \ldots, P_4R$, of heights $1, 3, 2, 2, 3$, and

$$K_{r,s}^{(q)} = (t^q) \cap \bigcap_{j=1}^{4} Q_j^q R = t^q I_{r,s}^q R = K_{r,s}^q.$$

Moreover $\sqrt{K_{r,s}} = tMR$ and support preservation holds. Its support complex is a cone and hence contractible, with $\beta_1 = 0$ over every field. A squarefree diagonal root exists exactly when $r = s$; for $r = s = \rho$, it is tMR, with weights $(1, 1, 1, 1, \rho, 1)$. Consequently the MPV hypotheses, support preservation, and $\beta_1 \leq 1$ together still do not imply the MPV conclusion.

Proof. Extension by t commutes with intersections of monomial ideals, and $(t^q) \cap LR = t^q LR$ for every ideal $L \subset S$. These identities give the decomposition and the power formulas. Irredundance follows from the witnesses in Theorem 3.1, multiplied by t, and the witness ac for the component (t). The radicals are incomparable. The generators are $tac, tbd, tabz^r, tcdz^s$; their supports are obtained by adjoining t to every original facet. This proves the cone and radical statements. Proposition 2.1 gives the exact representation boundary and weights. □

Coning preserves strong cycles. In fact, if $\mathcal{H}^+ = \{E \cup \{t\} : E \in \mathcal{H}\}$, then t cannot be a cycle vertex: it belongs to every cycle edge, contrary to the strong-cycle condition for length at least three. Deleting t from each edge therefore gives a length-preserving bijection of strong cycles, with inverse obtained by adjoining t. In particular balancedness is preserved. The coned generator clutter still contains the displayed strong triangle, with t adjoined to each edge, while its component clutter contains the second displayed triangle unchanged. The cone example therefore does not refute either balanced-support requirement.

4 The no-strong-triangle criterion

Theorem 4.1. *Let $I \subsetneq S$ be nonzero and satisfy the MPV hypotheses. Suppose that*

$$\mathcal{F}(I) = \mathcal{F}(\sqrt{I}) = \mathcal{H}$$

and that $\mathcal{H}$ has no strong triangle. Each support $F \in \mathcal{H}$ then carries exactly one minimal generator u_F. For each variable x_i, its exponent in u_F is independent of the facet F containing it. Let d_i be this common positive exponent, and put $d_i = 1$ for unused variables. Then

$$I = J^{\delta_d}, \qquad J = \sqrt{I} = (\prod_{x_i \in F} x_i : F \in \mathcal{H}),$$

and J is squarefree and Simis. In particular the conclusion holds when $\mathcal{H}$ is balanced.

We separate the established algebraic ingredients from the combinatorial step.

Lemma 4.2 (Bordoloi–Das–Kumar)**.** *Under the MPV hypotheses and support preservation:*

(i) every member of $\mathcal{F}(I)$ supports exactly one minimal generator;

(ii) if distinct $F, G \in \mathcal{F}(I)$ satisfy

$$\{E \in \mathcal{F}(I) : E \subseteq F \cup G\} = \{F, G\}, \tag{8}$$

then the exponents of every shared variable agree in u_F and u_G.

Part (i) is [BDK26a, Lemma 2.1], and part (ii) is [BDK26a, Proposition 2.2], with the hypotheses used here. The relevant induced complex is generated by facets contained in $F \cup G$.

For clarity, the witness underlying part (ii) is recalled. Write $u_F = x^a$, $u_G = x^b$, and put $m_i = \min(a_i, b_i)$, $M_i = \max(a_i, b_i)$ on $F \cap G$. If a shared exponent differs, set

$$f = \prod_{x_i \in F \cap G} x_i^{\max(2m_i, M_i)} \prod_{x_i \in F \setminus G} x_i^{a_i} \prod_{x_i \in G \setminus F} x_i^{b_i}. \tag{9}$$

Isolation and part (i) restrict possible products dividing f to $u_F^2, u_F u_G, u_G^2$. Private variables exclude the squares, while a mismatching shared coordinate excludes $u_F u_G$, since $\max(2m_i, M_i) <$

$m_i + M_i$. Thus $f \notin I^2$. In a pure-power component Q, choose generators dividing u_F and u_G. If they use the same variable, its doubled threshold divides f; otherwise their product divides f. Hence $f \in Q^2$ for every component, so $f \in I^{(2)}$, a contradiction. This is the existing isolated-pair obstruction, not an additional result.

Lemma 4.3 (Isolation of an unequal incidence)**.** *Let $\mathcal{H}$ be a clutter without a strong triangle. Fix a vertex z, put $\mathcal{H}_z = \{E \in \mathcal{H} : z \in E\}$, and let $\lambda : \mathcal{H}_z \to \Lambda$ be a nonconstant labeling. There are $F, G \in \mathcal{H}_z$ with $\lambda(F) \neq \lambda(G)$ satisfying (8), with $\mathcal{H}$ in place of $\mathcal{F}(I)$.*

Proof. Choose an unequal-label pair A, B minimizing $|A \cup B|$, and put

$$P = A \setminus B, \qquad Q = B \setminus A, \qquad D = A \cap B.$$

Suppose a third edge $C \subseteq A \cup B$ exists. Since the edges form an antichain, C meets both P and Q; choose $p \in C \cap P$ and $q \in C \cap Q$. If $d \in D \setminus C$, then

$$A \cap \{d, p, q\} = \{d, p\}, \quad C \cap \{d, p, q\} = \{p, q\}, \quad B \cap \{d, p, q\} = \{q, d\},$$

a strong triangle. Thus $D \subseteq C$, and in particular $z \in C$.

The clutter property also gives $A \not\subseteq C$ and $B \not\subseteq C$. As $D \subseteq C$, some vertex of P and some vertex of Q are absent from C. Hence

$$A \cup C \subsetneq A \cup B, \qquad B \cup C \subsetneq A \cup B.$$

Minimality forces both $\lambda(A) = \lambda(C)$ and $\lambda(B) = \lambda(C)$, contradicting the choice of A, B. No third edge exists. □

Proof of Theorem 4.1. Lemma 4.2(i) supplies $u_F = x^{\alpha(F)}$ for each $F \in \mathcal{H}$. For a used variable x_i, label the facets through it by $\lambda_i(F) = \alpha(F)_i$. If this labeling were nonconstant, Lemma 4.3 would give an isolated unequal-label pair, contrary to Lemma 4.2(ii). Thus every positive occurrence exponent at x_i is the same. Proposition 2.1 gives the representation and identifies $J = \sqrt{I}$. Equation (4) gives Simisness of J. Balancedness excludes all strong odd cycles, in particular strong triangles. □

The theorem is sufficient, not necessary; Section 6.2 gives represented Simis ideals whose generator clutters contain strong triangles. Nor does the theorem assert that every exact support-3 ideal has a triangle-free clutter. The comparison with [BDK26a] concerns a new support condition, not a replacement of every support-size result.

5 A sharp criterion on component supports

The next criterion uses the supports A_j of irreducible components, rather than the supports of minimal generators. Its algebraic input is the following three-variable obstruction.

Lemma 5.1 (Core square obstruction)**.** *In $k[y_0, y_1, y_2]$, let*

$$A = (y_0^{\lambda}, y_1^{\alpha}), \qquad B = (y_0^{\mu}, y_2^{\beta}), \qquad 0 < \lambda < \mu,$$

with $\alpha, \beta > 0$. Then $(A \cap B)^2 \subsetneq A^2 \cap B^2$. If $\gamma, \delta > 0$ and either

$$C = (y_1^{\gamma}, y_2^{\delta}) \quad \text{or} \quad C = (y_0^{\theta}, y_1^{\gamma}, y_2^{\delta}), \quad \theta > 0,$$

then $(A \cap B \cap C)^2 \subsetneq A^2 \cap B^2 \cap C^2$.

Proof. All witnesses below are monomials. For two components, put $M = \max(2\lambda, \mu)$ and $f = y_0^M y_2^\beta$. It belongs to $A^2 \cap B^2$. The only minimal intersection generators dividing it are y_0^μ and $y_0^\lambda y_2^\beta$. Their products fail to divide f, because

$$2\mu > M, \qquad \mu + \lambda > M, \qquad 2\beta > \beta.$$

Now suppose C is present, and write

$$\widehat{\alpha} = \max(\alpha, \gamma), \qquad \widehat{\beta} = \max(\beta, \delta), \qquad K = A \cap B \cap C.$$

First assume that C has no y_0-generator, or that $\theta > \mu$. The monomial

$$F = y_0^\mu y_1^{\widehat{\alpha}} y_2^{\widehat{\beta}}$$

belongs to $A^2 \cap B^2 \cap C^2$, using the products $y_0^\lambda y_1^\alpha$, $y_0^\mu y_2^\beta$, and $y_1^\gamma y_2^\delta$. If $F \notin K^2$, we are done. Otherwise two lcm generators of K have product dividing F. In the order (y_0, y_1, y_2), all candidates are among

$$\begin{aligned} &(\mu, \gamma, 0), \quad (\mu, \widehat{\alpha}, 0), \quad (\mu, 0, \delta), \quad (\lambda, 0, \widehat{\beta}), \\ &(0, \widehat{\alpha}, \beta), \quad (0, \alpha, \widehat{\beta}), \quad (\lambda, \gamma, \beta), \quad (\mu, \alpha, \delta). \end{aligned} \tag{10}$$

Any candidate with first coordinate μ cannot pair with one having positive first coordinate. Pairing it with either candidate having first coordinate zero exceeds $\widehat{\alpha}$ or $\widehat{\beta}$. Among the remaining candidates, a coordinate comparison excludes every pair except two copies of (λ, γ, β). Thus

$$2\lambda \le \mu, \qquad \widehat{\alpha} = \alpha \ge 2\gamma, \qquad \widehat{\beta} = \delta \ge 2\beta.$$

Set $G = y_0^\lambda y_1^\alpha y_2^\delta$. It lies in all three squares, using $y_0^\lambda y_1^\alpha$, $y_2^{2\beta}$, and $y_1^\gamma y_2^\delta$, respectively. The lcm generators that can divide G, after dropping a redundant candidate, have vectors

$$D = (\lambda, 0, \delta), \qquad E = (0, \alpha, \beta), \qquad T = (\lambda, \gamma, \beta).$$

Every product involving D exceeds the y_2-bound; $E + E$ and $E + T$ exceed the y_1-bound; $T + T$ exceeds the y_0-bound. Hence $G \notin K^2$.

It remains to consider the three-generator form with $\theta \le \mu$. Put

$$M = \max(2\lambda, \mu), \quad N = \begin{cases} \beta, & 2\theta \le M, \\ \widehat{\beta}, & 2\theta > M, \end{cases} \qquad H = y_0^M y_2^N.$$

Membership in A^2 and B^2 is immediate. For C^2, use $y_0^{2\theta}$ in the first case and $y_0^\theta y_2^\delta$ in the second. With $h = \max(\lambda, \theta)$, the lcm candidates dividing H are among

$$(\mu, 0, 0), \quad (\lambda, 0, \widehat{\beta}), \quad (\mu, 0, \delta), \quad (h, 0, \beta).$$

Any product involving first coordinate μ exceeds M. Products involving $(\lambda, 0, \widehat{\beta})$ and another candidate with positive third coordinate exceed N. Only two copies of $(h, 0, \beta)$ remain. If $2\theta \le M$, their third coordinate exceeds $N = \beta$; if $2\theta > M$, then $h = \theta$, and their first coordinate exceeds M. Thus $H \notin K^2$. □

Theorem 5.2 (At most one intervening component)**.** *Let $I \subsetneq S$ be nonzero, with no embedded primes and irredundant irreducible decomposition (1) having pairwise distinct radicals. For $k < \ell$, put*

$$N(k,\ell) = \{j \notin \{k,\ell\} : A_j \subseteq A_k \cup A_\ell\}.$$

Assume

$$|N(k,\ell)| \leq 1 \quad \text{whenever } A_k \cap A_\ell \neq \varnothing. \tag{11}$$

If $I^{(2)} = I^2$, then $I = J^{\delta_d}$ for a squarefree ideal $J \subset S$. More precisely, $J = \sqrt{I} = \bigcap_j P_j$, all positive occurrence exponents at each variable coincide, and the used weights are forced; unused weights are arbitrary. The prime intersection is irredundant.

Condition (11) holds automatically when every $|A_j| = 2$. The bound one is sharp: replacing it by two permits the nonrepresentable all-power Simis ideals $I_{r,s}$, $r \neq s$, of Theorem 3.1.

Proof. The structural hypotheses give $I^{(2)} = \bigcap_j Q_j^2$ and incomparability of the A_j. Suppose $a_{k,i} = \lambda < \mu = a_{\ell,i}$ for some shared variable. Put $A = A_k$, $B = A_\ell$, and $U = A \cup B$. Invert all variables outside U. Every component with support not contained in U becomes the unit ideal; by (11), only the components on A, B, and possibly one further support C, survive.

Contract to $R_U = k[x_u : u \in U]$. Polynomial extension by invertible outside variables preserves and reflects membership in these monomial ideals. Thus the assumed equality yields

$$(Q_A \cap Q_B)^2 = Q_A^2 \cap Q_B^2, \tag{12}$$

or the corresponding equality with Q_C included on both sides.

Set $y_0 = x_i$. If C is absent, choose y_1 from $A \setminus B$ and y_2 from $B \setminus A$. If C is present, its incomparability with A, B, together with $C \subseteq A \cup B$, permits the choices

$$y_1 \in (A \cap C) \setminus B, \qquad y_2 \in (B \cap C) \setminus A.$$

Restrict (12) to monomials supported on these three variables. The first two components become

$$(y_0^\lambda, y_1^\alpha), \qquad (y_0^\mu, y_2^\beta).$$

The possible third becomes (y_1^γ, y_2^δ) or $(y_0^\theta, y_1^\gamma, y_2^\delta)$, according as $y_0 \notin C$ or $y_0 \in C$. Restriction commutes with the intersections and squares in question: a monomial involving an omitted variable cannot divide a monomial in the smaller ring. Lemma 5.1 contradicts the restricted equality.

No exponent mismatch can therefore occur. Proposition 2.1 gives the root, its uniqueness, irredundance, and the weight statements.

For two distinct intersecting two-element supports $\{i,j\}, \{i,h\}$, the only further two-element support in their union is $\{j,h\}$. This proves the height-two assertion. Finally, $I_{r,s}$ has four components, so every pair has at most two intervenors. The mismatching pair with supports $\{a,b,z\}$ and $\{c,d,z\}$ has exactly the two intervenors $\{a,d\}$ and $\{b,c\}$. For $r \neq s$, Theorem 3.1 gives all-power Simisness and failure of representability. □

Corollary 5.3. *Under the structural and incidence assumptions of Theorem 5.2, if I is Simis, then its forced squarefree root is Simis. Hence the MPV conclusion holds.*

Proof. Apply Theorem 5.2 and (4). □

5.1 Comparison with the mismatch criterion

In the notation of Proposition 2.1, Bijender–Kumar prove the following [BK25, Notation 3.5, Set-up 4.1 and Theorem 4.7]. For an irredundant irreducible decomposition with distinct radicals, no embedded primes, and $\bigcup_j A_j = \{1, \dots, n\}$, assume

$$N(k, \ell) = \varnothing \qquad \text{for every } (k, \ell) \in X_I. \tag{13}$$

Then I is Simis if and only if it is a standard diagonal weighting of a squarefree Simis ideal. Their condition concerns only mismatching pairs; (11) concerns every intersecting pair but permits one intervenor. These antecedents should not be compared merely by comparing the numbers zero and one. The height-two case also occurs in [BK25, Theorems 4.9 and Corollary 4.10].

The distinct-radical convention cannot be omitted. Indeed,

$$(x, y)^2 = (x^2, y) \cap (x, y^2)$$

is an irredundant irreducible decomposition with equal radicals. The ideal has no embedded primes and is Simis, since every power is (x, y)-primary. There is no third component, so the intervenor condition is vacuous, but the unequal occurrence exponents preclude a squarefree diagonal root.

There is a second qualification. If I has a squarefree diagonal representation, Proposition 2.1 gives $X_I = \varnothing$. Consequently (13) is automatically satisfied. If unused variables are present, restrict to the polynomial ring on $\bigcup_j A_j$, apply the criterion there, and extend back; monomial membership and the symbolic-power formulas show that Simisness is unchanged. Thus represented Simis examples satisfying the MPV hypotheses cannot separate the successful class of [BK25, Theorem 4.7]. The distinction supplied by Theorem 4.1 is a support condition proving exponent coherence before the representation is known.

6 Examples and the limits of the criteria

6.1 Balanced support four beyond forests

We use the established implication that a balanced clutter is Mengerian [FHO74], and the equivalence between the Mengerian property and equality of all ordinary and symbolic powers of its squarefree facet ideal [HHTZ08, Section 2 and Corollary 1.6]. In particular a balanced squarefree facet ideal is Simis and normally torsion-free. Here normally torsion-free means $\operatorname{Ass}(S/L^q) \subseteq \operatorname{Ass}(S/L)$ for every $q \geq 1$. These results supply the squarefree Simis input to the following family.

Proposition 6.1 (Even-cycle family)**.** *Fix $m \geq 2$, put $n = 2m$, and read subscripts modulo n. On the $3n = 6m$ variables*

$$V_m = \{x_1, \dots, x_n, y_1, \dots, y_n, z_1, \dots, z_n\},$$

set

$$E_i = \{x_i, x_{i+1}, y_i, z_i\}, \quad \mathcal{H}_m = \{E_1, \dots, E_n\}, \quad \Delta_m = \langle \mathcal{H}_m \rangle.$$

In $S_m = k[V_m]$, let

$$J_m = \Big(\prod_{v \in E_i} v : 1 \leq i \leq n\Big), \qquad I_{m,d} = \Big(\prod_{v \in E_i} v^{d_v} : 1 \leq i \leq n\Big),$$

where $d \in \mathbb{Z}_{>0}^{V_m}$ is arbitrary. Then:

(i) the displayed lists are minimal generating sets, $\sqrt{I_{m,d}} = J_m$*, and* $\mathcal{F}(I_{m,d}) = \mathcal{F}(J_m) = \mathcal{H}_m$*;*

(ii) writing $\mathcal{C}_m = b(\mathcal{H}_m)$*, the decompositions*

$$J_m = \bigcap_{C\in\mathcal{C}_m} (v : v \in C), \qquad I_{m,d} = \bigcap_{C\in\mathcal{C}_m} (v^{d_v} : v \in C)$$

are irredundant, with distinct incomparable radicals;

(iii) $I_{m,d} = J_m^{\delta_d}$*, both ideals are Simis and normally torsion-free, and*

$$\operatorname{Ass}(S_m/I_{m,d}^q) = \{(v : v \in C) : C \in \mathcal{C}_m\} \quad (q \geq 1);$$

(iv) $\mathcal{H}_m$ *is balanced and 4-uniform, but* Δ_m *is not a forest. Every nonempty proper facet-generated subcomplex is a forest.*

Thus every $d \neq \mathbf{1}$ *gives a nonsquarefree ideal satisfying the MPV hypotheses and the balanced-support criterion, but outside the forest and exact-support-3 classes of [BDK26a]. It also satisfies the condition that deleting any one facet leaves a forest. These examples have no generator of support size at most three.*

Proof. The vertices y_i, z_i are private to E_i, and x_i belongs exactly to E_{i-1}, E_i. Consecutive facets meet in one x-variable, and nonconsecutive facets are disjoint. Any strong-cycle vertex must be an x_i, so the cycle follows the ordinary cyclic incidence pattern on these variables. A simple closed walk in this cycle graph uses the entire cycle; its length is $n = 2m$. Hence every strong cycle has even length and $\mathcal{H}_m$ is balanced.

No E_i is a leaf: a joint would have to contain both intersections $\{x_i\}$ with E_{i-1} and $\{x_{i+1}\}$ with E_{i+1}, but no other facet does. On deleting at least one facet, the remaining cyclic indices form disjoint paths. Every nonempty subfamily has a path endpoint, which is a leaf, or an isolated facet, which is also a leaf. Thus every proper facet-generated subcomplex is a forest.

The private vertices show that the displayed generators are pairwise nondivisible, before and after weighting. The radical and weighting assertions follow directly. The minimal-cover decomposition of J_m gives the stated primes. To check irredundance explicitly, for a fixed cover C choose $v_D \in D \setminus C$ for each other minimal cover D. Then $\prod_{D\neq C} v_D$ belongs to every other prime and not the prime on C. Weighting this witness proves irredundance of the weighted decomposition. Minimal covers form an antichain, so the radicals are distinct and incomparable.

Balancedness gives Simisness and normal torsion-freeness of J_m by the cited results. Equation (4) gives Simisness of $I_{m,d}$. Since

$$I_{m,d}^q = (J_m^q)^{\delta_d},$$

preservation of associated primes under weighting gives the assertion for every power. Finally, every variable occurs in a facet; if $d \neq \mathbf{1}$, some minimal generator has an exponent exceeding one, so the weighted ideal is not squarefree. □

This is an infinite family of ambient examples. For instance, take $d_{y_1} = 2$ and all other weights one. The rings have dimensions $6m$, so the pairs obtained as m varies are pairwise nonisomorphic. The strict comparison is with the named forest and support-size antecedents, not with every previously known sufficient condition. The family does not establish a general theorem from the one-facet-deletion condition alone.

6.2 Strong triangles in represented ideals

Proposition 6.2 (Disjoint-block family)**.** *Let $B_1, \dots, B_m$ be disjoint nonempty finite sets of variables, $S = k[\bigsqcup_h B_h]$, and $a = (a_z)$ a positive integer weighting. Put*

$$P_h = (z : z \in B_h), \quad Q_h = (z^{a_z} : z \in B_h), \quad I_a = \bigcap_h Q_h, \quad J = \bigcap_h P_h.$$

The displayed decompositions are irredundant, with distinct incomparable radicals, and $\operatorname{ht} P_h = |B_h|$. *For every $q \geq 1$,*

$$I_a^q = I_a^{(q)} = \bigcap_h Q_h^q, \qquad J^q = J^{(q)} = \bigcap_h P_h^q,$$

and the associated primes of both powers are exactly $\{P_1, \dots, P_m\}$. Moreover,

$$\mathcal{G}(J) = \{z_1 \cdots z_m : z_h \in B_h\}, \qquad \mathcal{G}(I_a) = \{z_1^{a_{z_1}} \cdots z_m^{a_{z_m}} : z_h \in B_h\}.$$

Thus support preservation holds, $\sqrt{I_a} = J$, and $I_a = J^{\delta_a}$, with unique root and forced weights on every variable. The ideal is nonsquarefree exactly when $a \neq \mathbf{1}$.

If at least two blocks have size at least two, the common facet complex has no leaf. If at least three blocks have size at least two, the common generator clutter contains a strong triangle. Nevertheless $N(k, \ell) = \varnothing$ for every component pair, and $X_{I_a} = \varnothing$.

Proof. For monomial ideals on disjoint blocks, intersection equals product: a monomial in every factor is divisible by a product of one generator from each block. Consequently

$$I_a = \prod_h Q_h, \qquad I_a^q = \prod_h Q_h^q = \bigcap_h Q_h^q.$$

Each Q_h^q is P_h-primary. One way to see the contraction assertion used here is to write the quotient as $B \otimes_k k[Z]$, where B is the finite-dimensional local quotient on B_h. The powers of its nilpotent maximal ideal give a finite filtration with free $k[Z]$-quotients. Every polynomial outside P_h acts injectively on these quotients, hence on the quotient module. Thus localization at P_h contracts Q_h^q to itself.

Choose $z_\ell \in B_\ell$. The witness $\prod_{\ell \neq h} z_\ell^{q a_{z_\ell}}$ belongs to every Q_ℓ^q except Q_h^q. This proves irredundance for every power, identifies associated primes, and, with localization, gives the symbolic-power formulas. Taking all weights one gives the assertions for J.

Expanding the product gives the displayed generators. Different transversals are incomparable, so these lists are minimal. The radical, weighting, and uniqueness assertions follow from Proposition 2.1. Every variable occurs in a transversal, so weights are forced and nonsquarefreeness is equivalent to $a \neq \mathbf{1}$.

Suppose two blocks have size at least two. For any facet F and proposed joint $G \neq F$, choose a block where their entries differ, and a second block with at least two entries. Form H by changing only the entry of F in the second block. The entry in the first block belongs to $F \cap H$ but not G; thus G is not a joint. No facet is a leaf.

For the strong triangle, choose distinct $p_i, q_i \in B_i$, $i = 1, 2, 3$, and fixed entries in the remaining blocks. The facets with first three entries

$$(p_1, p_2, q_3), \qquad (q_1, p_2, p_3), \qquad (p_1, q_2, p_3)$$

meet $\{p_1, p_2, p_3\}$ in exactly its three two-element subsets. Finally, disjointness of blocks gives both $N(k, \ell) = \varnothing$ and $X_{I_a} = \varnothing$. □

Example 6.3 (A support-4 specialization)**.** Take

$$B_1 = \{u_1, v_1, w_1\}, \quad B_2 = \{u_2, v_2\}, \quad B_3 = \{u_3, v_3\}, \quad B_4 = \{u_4, v_4\},$$

and set every weight equal to two. There are $3 \cdot 2 \cdot 2 \cdot 2 = 24$ minimal generators, each of support size four, and component heights $(3, 2, 2, 2)$. The ideals are

$$J = (u_1, v_1, w_1)(u_2, v_2)(u_3, v_3)(u_4, v_4),$$

$$I_a = (u_1^2, v_1^2, w_1^2)(u_2^2, v_2^2)(u_3^2, v_3^2)(u_4^2, v_4^2).$$

The facets

$$\{u_1, u_2, v_3, u_4\}, \quad \{v_1, u_2, u_3, u_4\}, \quad \{u_1, v_2, u_3, u_4\}$$

form a strong triangle on u_1, u_2, u_3. The ideal is nonsquarefree, satisfies the MPV conclusion and the component criterion, but fails the forest, balanced, no-strong-triangle, support-at-most-three, and component-height-at-most-two conditions. This proves nonnecessity of the no-strong-triangle condition. Since $X_{I_a} = \varnothing$, it is not a separating example for the mismatch criterion of [BK25].

A Use of generative AI and provenance

The research reported in this paper was carried out with the assistance of Pharos (website: github.com/frenzymath/Pharos), an automated mathematical reasoning system built on Rethlas [Ju+26]. Two projects were initiated on 18 October 2026 and 20 October 2026, respectively. The first project was supplied with the Méndez–Vaz Pinto–Villarreal conjecture and relevant references and produced a counterexample to the conjecture. The second project was supplied with this counterexample and tasked with finding additional hypotheses under which the conjectured conclusion holds, including cases beyond the positive classes treated in [BDK26a]. The present manuscript was prepared from the drafts produced by these two projects.

Each project initially comprised a coordinator and five theorem-producing workers. Pharos allows the coordinator to recruit additional helper and verifier agents autonomously, without human intervention in these operational decisions. All coordinators, workers, helpers and verifiers used internal model of Hy.

No additional mathematical guidance was supplied for the project after its initial input. The only subsequent user intervention was a message identifying some mistakes after the first draft had been produced.

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